Program: **Diploma in Microelectronics**	
Course Code: **5499B**	Course Title: **VLSI Design Lab**
Semester: **5**	Credits: **1.5**
Course Category: **Program Elective**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- To study EDA tools used for VLSI Design (Synopsis, Mentor Graphics, Cadence etc.)

Course Prerequisites:

Topic	Course code	Course name	Semester
Combinational & Sequential circuits		Digital Electronics	3
CMOS		VLSI Design	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Implement the layout of various CMOS logic Gates	15	Applying
CO2	Implement layout of various CMOS combinational Circuit	15	Applying
CO3	Implement layout of various CMOS Sequential Circuit	12	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	3		3	2
CO2	3	3	3	3		3	2
CO3	3	3	3	3		3	2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Implement various CMOS logic Gates**		
M1.01	Simulate and Synthesize the layout of CMOS inverter	3	Applying
M1.02	Simulate and Synthesize the layout of NOR Gate	3	Applying
M1.03	Simulate and Synthesize the layout of NAND Gate	3	Applying
M1.04	Simulate and Synthesize the layout of AND/OR Gate	3	Applying
M1.05	Simulate and Synthesize the layout of Design and simulation of XOR Gate	3	Applying
CO2	**Implement various CMOS combinational Circuit**		
M2.01	Simulate and Synthesize the layout of Half Adder	3	Applying
M2.02	Simulate and Synthesize the layout of Half Subtractor	3	Applying
M2.03	Simulate and Synthesize the layout of Full Adder	4.5	Applying
M2.04	Simulate and Synthesize the layout of Full Subtractor	4.5	Applying
	Lab Exam	1.5	
CO3	**Implement various CMOS Sequential Circuit**		
M3.01	Simulate and Synthesize the layout of D Flip flop	3	Applying
M3.02	Simulate and Synthesize the layout of T Flip flop	3	Applying

M3.03	Simulate and Synthesize the layout of JK Flip flop	3	Applying
M3.04	Simulate and Synthesize the layout of SR Flip flop	3	Applying
	Lab Exam	1.5	

****Suggested Open Ended Projects**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can-do open-ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation and a score of 15 marks. Students should prepare a separate report on open ended experiment of their choice.)

Example:

1. Implement the layout of 2-bit counter.

2. Simulate the layout of sequence generator which can count (0, 2, 4…)

Text /Reference:

T/R	Book Title/Author
T1	CMOS: Circuit Design, Layout, and Simulation, R. Jacob Baker, 2019, IEEE Press on Microelectronics systems
T2	Digital VLSI Chip Design with Cadence and Synopsys CAD Tools, Erik Brunvand, 2009
R1	Analog IC Design, Behzad, Razavi, Mc Graw Hill,
R2	Basic VLSI Design, Puknell D A, 1995, PHI

Online Resources:

Sl. No	Website Link
1	https://www.eda-solutions.com/products/tanner-designer/
2	https://www.montana.edu/aolson/eele414/information/Guide_to_TannerEDA_for_VLSI.pdf